

Callum T. Donnan

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Research Interests

Galaxy formation & evolution, first stars & galaxies, cosmic reionization, stellar populations, large-scale structure, galaxy-halo connection

Education

PhD Astrophysics, University of Edinburgh, Edinburgh, UK 2021 - 2025

- Thesis: *"Exploring the formation of the first galaxies with VISTA and JWST"*
- Supervisors: James Dunlop, Ross McLure, Derek McLeod

MPhys Astrophysics, University of St Andrews, St Andrews, UK 2016 - 2021

- Thesis: *"The gas content of galaxies in the cosmic web"*
- Supervisor: Rita Tojeiro

Research Positions

Post Doctoral Research Associate, NOIRLab, Tucson, AZ, USA 2025 - Present

Summer Research Student, University of St Andrews, St Andrews, UK 2021

Institute of Astronomy Summer Research Programme, University of Cambridge, Cambridge, UK 2020

Summer Research Student, University of St Andrews, St Andrews, UK 2019

Awards & Grants

JWST Cycle 4 PI GO 6954, ~ \$95k 2025

MPhys Thesis Prize for best MPhys astrophysics thesis 2021

Dean's List (awarded for averaging a first class across the academic year.) 2016 - 2021

Cormack Vacation Scholarship from the Royal Society of Edinburgh 2019

School of Physics & Astronomy Student Staff Council Vacation Award 2019

Medal (Astronomy and Astrophysics First & Second Level) 2016-2018

Margaret Stewart Prize (Awarded to the best First Year student in Astronomy and Astrophysics) 2016 - 2017

Telescope Programs

PI, Spectroscopic confirmation of the most robust $z=15$ galaxy candidate, GO 6954, 6 hours JWST Cycle 4

Co-I, The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS), GO 6368, 194 hours, PI: M. Dickinson JWST Cycle 3

Co-I, Dead or alive? Unveiling the nature of massive galaxies in the early Universe, GO 5545, 10 hours, PI: L. Barrufet JWST Cycle 3

Co-I, EXCELS: The Early eXtragalactic Continuum and Emission Line Survey, GO 3543, 72 hours, PI: A. Carnall & F. Cullen JWST Cycle 2

Co-I, Characterizing true $z > 12$ galaxies and $z \sim 4 - 7$ interlopers in preparation for JWST Cycle 2, DD 2750, 8 hours, PI: P. Arrabal Haro JWST DDT

Co-I, The AURORA Survey: First Direct Metallicity Calibrations at High Redshift , GO 1914, 63 hours, PI: A. Shapley & R. Sanders	JWST Cycle 1
Co-I, PRIMER: Public IMaging for Extragalactic Research , GO 1837, 196 hours, PI: J. Dunlop	JWST Cycle 1

Supervision, Teaching & Outreach

Supervisor , undergraduate summer research project	2024
Teaching Assistant , Physics 1A, Discovering Astronomy, Observational Astronomy	2021-2024
Royal Observatory open day , JWST presentation to members of the public	2023
High School Outreach , presentations to high school students on astronomy	2018-2023

Press

Physics magazine , JWST Sees More Galaxies than Expected, link	2024
Physicsworld , JWST spectrometer refines redshifts of distant galaxies, link	2023
Le Parisein , Distant galaxies: the James Webb Telescope defies all odds, link	2023
Nature News , Four revelations from the Webb telescope about distant galaxies, link	2022
BBC News , Scottish astronomers push James Webb deeper back in time, link	2022
VICE , A Cosmic Web Connecting the Universe Shapes Dark Matter in Galaxies, Study Finds, link	2022

Publications (41, 7 as first author, h-index: 23, citations: 2540)

First Author:

1. Donnan, C.T. , Dickinson, M., Taylor, A. J., et al., Correcting the galaxy stellar mass function (GSMF) at $z \geq 5$ using JWST CAPERS	In prep.
2. Donnan, C.T. , Dickinson, M., Taylor, A. J., et al., arXiv:2507.10518, Very bright, very blue, and very red: JWST CAPERS analysis of highly luminous galaxies with extreme UV slopes at $z = 10$	2025
3. Donnan, C.T. , Dunlop, J. S., McLure, R. J., et al., <i>MNRAS</i> , 539, 2409, No evidence (yet) for increased star-formation efficiency at $z = 6 - 13$	2025
4. Donnan, C.T. , McLure, R. J., Dunlop, J. S., et al., <i>MNRAS</i> , 533, 3222, JWST PRIMER: A new multi-field determination of the evolving galaxy UV luminosity function at redshifts $z \simeq 9 - 15$	2024
5. Donnan, C.T. , McLeod, D. J., McLure R. J., et al., <i>MNRAS</i> , 520, 4554, The abundance of $z \geq 10$ galaxy candidates in the HUDF using deep JWST NIRCам medium-band imaging.	2023
6. Donnan, C.T. , McLeod, D. J., Dunlop J. S., et al., <i>MNRAS</i> , 518, 6011, The evolution of the galaxy UV luminosity function at redshifts $z \simeq 8 - 15$ from deep JWST and ground-based near-infrared imaging.	2023
7. Donnan, C.T. , Tojeiro, R., Kraljic, K., <i>Nature Astronomy</i> , 6, 599, The role of the cosmic web in the scatter of the galaxy stellar mass - gas metallicity relation.	2022

Second Author:

8. McLeod, D. J., Donnan, C.T. , McLure R. J., et al., <i>MNRAS</i> , 527, 5004, The galaxy UV luminosity function at $z \simeq 11$ from a suite of public JWST ERS, ERO, and Cycle-1 programs	2024
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Co-Author:

9. Napolitano, L., Pentericci, L., Dickinson, M., et al. arXiv:2508.14171, Lyα visibility from $z = 4.5$ to 11 in the UDS field: evidence for a high neutral hydrogen fraction and small ionized bubbles at $z \sim 7$	2025
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10. Sanders, R. L., Shapley, A. E., et al. *arXiv:2508.10099*, [The AURORA Survey: High-Redshift Empirical Metallicity Calibrations from Electron Temperature Measurements at \$z = 2 - 10\$](#) 2025
11. Barrufet, L., Dunlop, J. S., Begley, R., et al. *arXiv:2508.05740*, [Strength in Numbers: Red Galaxies Bolster the Cosmic Star Formation Rate Density at \$z > 3\$](#) 2025
12. Taylor, A. J., Kokorev, V., Kocevski, D. D., et al. *ApJL*, 989, L7, [CAPERS-LRD-z9: A Gas-enshrouded Little Red Dot Hosting a Broad-line Active Galactic Nucleus at \$z = 9.288\$](#) 2025
13. Sanders, R. L., Shapley, A. E., Topping, M. W., et al. *ApJ*, 989, 209, [The AURORA Survey: The Nebular Attenuation Curve of a Galaxy at \$z = 4.41\$ from Ultraviolet to Near-infrared Wavelengths](#) 2025
14. Berg, D. A., Sanders, R. L., Shapley, A. E., et al. *arXiv:2507.17057*, [The AURORA Survey: Robust Helium Abundances at High Redshift Reveal A Subpopulation of Helium-Enhanced Galaxies in the Early Universe](#) 2025
15. Hamadouche, M. L., McLure, R. J., Carnall, A. C., et al. *MNRAS*, 541, 463, [JWST PRIMER: strong evidence for the environmental quenching of low-mass galaxies out to \$z \simeq 2\$](#) 2025
16. Arellano-Córdova, K. Z., Cullen, F., Carnall, A. C., et al. *MNRAS*, 540, 2991, [The JWST EXCELS survey: direct estimates of C, N, and O abundances in two relatively metal-rich galaxies at \$z \simeq 5\$](#) 2025
17. Cullen, F., Carnall, A. C., Scholte, D., et al. *MNRAS*, 540, 2176, [The JWST EXCELS survey: an extremely metal-poor galaxy at \$z = 8.271\$ hosting an unusual population of massive stars](#) 2025
18. Kokorev, V., Chávez Ortiz, Óscar A., et al. *ApJL*, 988, L10, [CAPERS Observations of Two UV-bright Galaxies at \$z > 10\$. More Evidence for Bursting Star Formation in the Early Universe](#) 2025
19. Scholte, D., Cullen, F., Carnall, A. C., et al. *MNRAS*, 540, 1800, [The JWST EXCELS survey: probing strong-line diagnostics and the chemical evolution of galaxies over cosmic time using \$T_e\$ -metallicities](#) 2025
20. Taylor, A. J., Finkelstein, S. L., Kocevski, D. D., et al. *ApJ*, 986, 165, [Broad-Line AGN at \$3.5 < z < 6\$: The Black Hole Mass Function and a Connection with Little Red Dots](#) 2025
21. Kocevski, D. D., Finkelstein, A. L., Barro G., et al., *ApJ*, 986, 126, [The Rise of Faint, Red AGN at \$z > 4\$: A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields](#) 2025
22. Lines, N. E. P., Bowler, R. A. A., Adams, N. J., et al., *MNRAS*, 539, 2685, [JWST PRIMER: a lack of outshining in four normal \$z = 4-6\$ galaxies from the ALMA-CRISTAL Survey](#) 2025
23. Liu, F.-Y., Dunlop, J. S., McLure, R. J., et al., *arXiv:2503.07774*, [JWST PRIMER: A deep JWST study of all ALMA-detected galaxies in PRIMER COSMOS – dust-obscured star-formation history back to \$z \simeq 7\$](#) 2025
24. Begley, R., McLure, R. J., Cullen, F., et al. *MNRAS*, 537, 3245, [The evolution of \$\[\text{OIII}\]+\text{H}\beta\$ equivalent width from \$z \simeq 3 - 8\$: implications for the production and escape of ionizing photons during reionization](#) 2025
25. Shapley, A. E., Sanders, R. L., Topping, M. W., et al. *ApJ*, 981, 167, [The AURORA Survey: An Extraordinarily Mature, Star-forming Galaxy at \$z \sim 7\$](#) 2025
26. Stanton, T. M., Cullen, F., Carnall, A. C., et al. *MNRAS*, 537, 1735, [The JWST EXCELS survey: tracing the chemical enrichment pathways of high-redshift star-forming galaxies with O, Ar, and Ne abundances](#) 2025
27. Shapley, A. E., Sanders, R. L., Topping, M. W., et al., *ApJ*, 980, 242, [The AURORA Survey: A New Era of Emission-line Diagrams with JWST/NIRSpec](#) 2025
28. Leung, G. C. K., Finkelstein, S. L., Pérez-González, P. G., et al., *arXiv:2411.12005*, [Exploring the Nature of Little Red Dots: Constraints on AGN and Stellar Contributions from PRIMER MIRI Imaging](#) 2024

29. Merlin, E., Santini, P., Paris, D, et al., *arXiv:2409.00169*, **ASTRODEEP-JWST: NIRCam-HST multiband photometry and redshifts for half a million sources in six extragalactic deep fields** 2024
30. Carnall, A. C., Cullen, F., McLure, R. J., et al., *MNRAS*, 534, 325, **The JWST EXCELS survey: Too much, too young, too fast? Ultra-massive quiescent galaxies at $3 < z < 5$** 2024
31. Stanton, T. M., Cullen, F., McLure, R. J., et al., *MNRAS*, 532, 3102, **The NIRVANDELS Survey: the stellar and gas-phase mass-metallicity relations of star-forming galaxies at $z = 3.5$** 2024
33. Varadaraj, R. G., Bowler, R. A. A., Jarvis, M. J., et al., *MNRAS*, 533, 3724, **The sizes of bright Lyman-break galaxies at $z \simeq 3 - 5$ with JWST PRIMER** 2024
32. Weibel, A., Oesch, P. A., Barrufet, L., et al., *MNRAS*, 533, 1808, **Galaxy Build-up in the first 1.5 Gyr of Cosmic History: Insights from the Stellar Mass Function at $z \sim 4 - 9$ from JWST NIRCam Observations** 2024
34. Begley, R., Cullen, F., McLure, R.J., *MNRAS*, 527, 4040, **Connecting the escape fraction of Lyman-alpha and Lyman-continuum photons in star-forming galaxies at $z \simeq 4 - 5$** 2024
35. Cullen, F., McLeod, D. J., McLure, R.J., *MNRAS*, 531, 997, **Evidence for the emergence of dust-free stellar populations at $z > 10$** 2024
36. Arrabal Haro, P., Dickinson, M., Finkelstein, S. L., *Nature*, 622, 707, **Confirmation and refutation of very luminous galaxies in the early Universe** 2023
37. Carnall, A. C., McLure, R. J., Dunlop, J. S., *Nature*, 619, 716, **A massive quiescent galaxy at redshift 4.658** 2023
38. Hamadouche, M. L., Carnall, A. C., McLure, R. J., *MNRAS*, 521, 5400, **The connection between stellar mass, age, and quenching time-scale in massive quiescent galaxies at $z \simeq 1$** 2023
39. Carnall, A. C., McLeod, D. J., McLure, R. J., *MNRAS*, 520, 3974, **A surprising abundance of massive quiescent galaxies at $3 < z < 5$ in the first data from JWST CEERS** 2023
40. Cullen, F., McLure, R. J., McLeod, D. J., *MNRAS*, 520, 14, **The ultraviolet continuum slopes (β) of galaxies at $z \simeq 8 - 16$ from JWST and ground-based near-infrared imaging** 2023
41. Carnall, A. C., Begley, R., McLeod, D. J., *MNRAS*, 518, 45, **A first look at the SMACS0723 JWST ERO: spectroscopic redshifts, stellar masses, and star-formation histories** 2023